

MERN STACK

DAY – 9

4 July 2026

JAVASCRIPT

JavaScript is a versatile, dynamically typed programming language that brings life to web pages by making them interactive. It is used for building interactive web applications and supports both client-side and server-side development.

- **Interpreted language:** Code is executed line by line.
- **Dynamically typed:** Variable types are determined at runtime.
- **Single-threaded:** Executes one task at a time (but supports asynchronous operations).

Adding JavaScript in an HTML Document

JavaScript is used to make web pages interactive and dynamic. It is added to an HTML document using the `<script>` tag.

Ways to add JavaScript:

1. **Inline JavaScript** – Written directly inside HTML elements using event attributes like `onclick`.
2. **Internal JavaScript** – Written inside the `<script>` tag within the HTML file (in the `<head>` or `<body>` section).
3. **External JavaScript** – Written in a separate `.js` file and linked using `<script src="script.js"></script>`.

Advantages of External JavaScript:

- Improves code organization and readability.
- Allows code reuse across multiple pages.
- Makes maintenance easier.
- Can improve page loading speed through browser caching.

console.log()

The console.log() function logs general information to the console. This is one of the most commonly used methods for debugging in JavaScript.

```
console.log("Hello, World!");
```

JavaScript Variables

JavaScript variables are used to store data values. Variables can be declared using var, let, or const. Since JavaScript is dynamically typed, you do not need to specify a data type.

Types of Variables:

- var – Function-scoped and can be redeclared. Mostly used in older JavaScript.
- let – Block-scoped and can be updated but not redeclared in the same scope.
- const – Block-scoped and cannot be reassigned after initialization.

```
let a=10;  
a=20;  
const b=20;  
var c=10;  
var c=20;
```

JavaScript Operators

JavaScript operators are special symbols or keywords used to perform operations on variables and values. They are essential for calculations, comparisons, assigning values, and controlling program logic.

Types of JavaScript Operators:

- Arithmetic Operators: Used to perform mathematical calculations such as addition (+), subtraction (-), multiplication (*), division (/), and modulus (%).
- Assignment Operators: Used to assign values to variables or update existing values. Examples include =, +=, -=, *=, and /=.
- Comparison Operators: Compare two values and return either true or false. Common operators are ==, ===, !=, >, <, >=, and <=.

- Logical Operators: Combine or reverse conditions in decision-making. They include && (AND), || (OR), and ! (NOT).
- Unary Operators: Work with a single operand. Examples include ++ (increment), -- (decrement), typeof (returns data type), and delete (removes an object property).
- String Operators: The + operator joins two or more strings, and += appends text to an existing string.

```
<script>
  let a = 10;
  let b = 20;

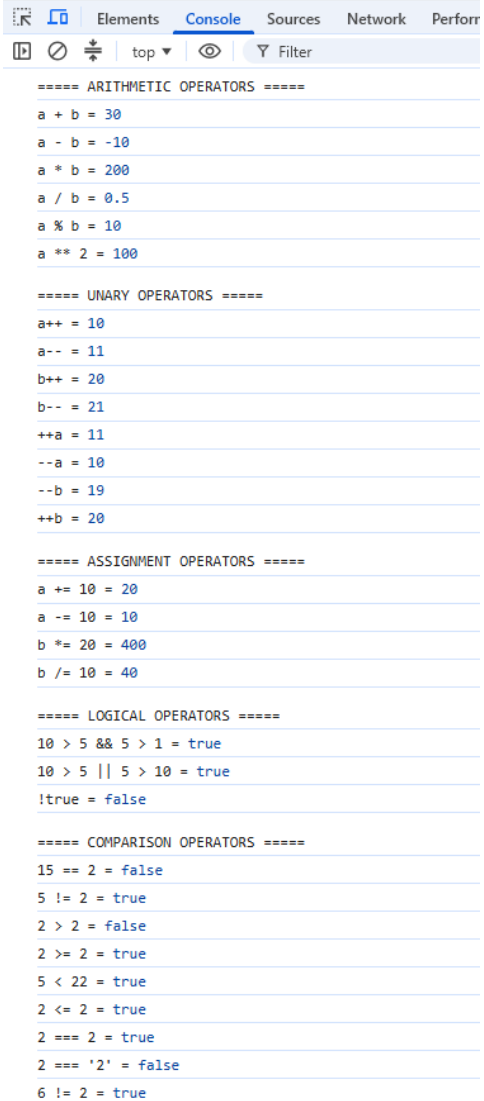
  console.log("==== ARITHMETIC OPERATORS =====");
  console.log("a + b =", a + b);
  console.log("a - b =", a - b);
  console.log("a * b =", a * b);
  console.log("a / b =", a / b);
  console.log("a % b =", a % b);
  console.log("a ** 2 =", a ** 2);

  console.log("\n==== UNARY OPERATORS =====");
  console.log("a++ =", a++);
  console.log("a-- =", a--);
  console.log("b++ =", b++);
  console.log("b-- =", b--);
  console.log("++a =", ++a);
  console.log("--a =", --a);
  console.log("--b =", --b);
  console.log("++b =", ++b);

  console.log("\n==== ASSIGNMENT OPERATORS =====");
  console.log("a += 10 =", (a += 10));
  console.log("a -= 10 =", (a -= 10));
  console.log("b *= 20 =", (b *= 20));
  console.log("b /= 10 =", (b /= 10));

  console.log("\n==== LOGICAL OPERATORS =====");
  console.log("10 > 5 && 5 > 1 =", 10 > 5 && 5 > 1);
  console.log("10 > 5 || 5 > 10 =", 10 > 5 || 5 > 10);
  console.log("!true =", !true);

  console.log("\n==== COMPARISON OPERATORS =====");
  console.log("15 == 2 =", 15 == 2);
  console.log("5 != 2 =", 5 != 2);
  console.log("2 > 2 =", 2 > 2);
  console.log("2 >= 2 =", 2 >= 2);
  console.log("5 < 22 =", 5 < 22);
  console.log("2 <= 2 =", 2 <= 2);
  console.log("2 === 2 =", 2 === 2);
  console.log("2 === '2' =", 2 === "2");
  console.log("6 != 2 =", 6 != 2);
</script>
```



```
==== ARITHMETIC OPERATORS =====
a + b = 30
a - b = -10
a * b = 200
a / b = 0.5
a % b = 10
a ** 2 = 100

==== UNARY OPERATORS =====
a++ = 10
a-- = 11
b++ = 20
b-- = 21
++a = 11
--a = 10
--b = 19
++b = 20

==== ASSIGNMENT OPERATORS =====
a += 10 = 20
a -= 10 = 10
b *= 20 = 400
b /= 10 = 40

==== LOGICAL OPERATORS =====
10 > 5 && 5 > 1 = true
10 > 5 || 5 > 10 = true
!true = false

==== COMPARISON OPERATORS =====
15 == 2 = false
5 != 2 = true
2 > 2 = false
2 >= 2 = true
5 < 22 = true
2 <= 2 = true
2 === 2 = true
2 === '2' = false
6 != 2 = true
```

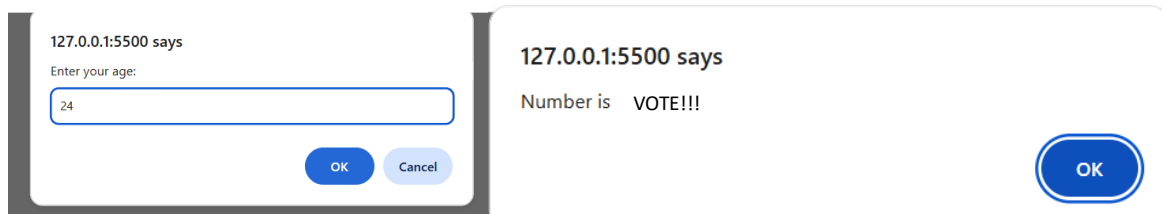
Conditional Statements in JavaScript

Conditional statements in JavaScript are used to make decisions in a program based on conditions. They control the flow of execution by running specific blocks of code depending on whether a condition is true or false.

Types of Conditional Statements:

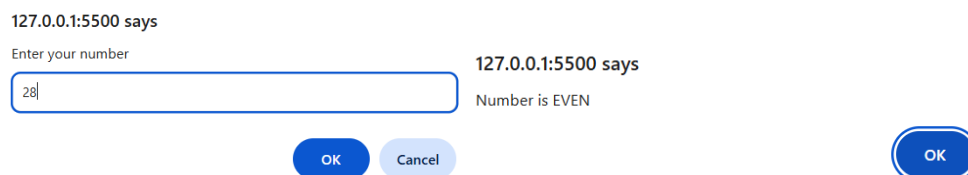
- **if statement:**
Executes a block of code only if the given condition is true.
- **if...else statement:**
Executes one block if the condition is true, otherwise executes another block.
- **if...else if...else statement:**
Used to check multiple conditions one by one. The first true condition's block is executed.
- **switch statement:**
Used to compare a single variable against multiple possible values and execute the matching case.

```
let age=prompt("Enter your age: ");
if(age>=18){
    alert("You can VOTE!!!");
}else{
    alert("You can't VOTE!!!");
}
```



The screenshot shows a web browser window with a prompt box titled "127.0.0.1:5500 says" containing the text "Enter your age:". The input field contains the number "24". Below the input field are "OK" and "Cancel" buttons. To the right, an alert box titled "127.0.0.1:5500 says" displays the message "Number is VOTE!!!".

```
let number=prompt("Enter your number");
if(number%2==0){
    alert("Number is EVEN");
}else{
    alert("Number is ODD");
}
```



The screenshot shows a web browser window with a prompt box titled "127.0.0.1:5500 says" containing the text "Enter your number". The input field contains the number "28". Below the input field are "OK" and "Cancel" buttons. To the right, an alert box titled "127.0.0.1:5500 says" displays the message "Number is EVEN".

parseInt() in JavaScript

- parseInt() is used to convert a string into an integer (whole number).
- It ignores decimal values and keeps only the integer part.
- If the value is not a number, it returns NaN.

```
<SCRIPT>
let a=parseInt(prompt("Enter a: "));
let b=parseInt(prompt("Enter b: "));
console.log("ARITHMETIC OPERATORS");
console.log("a + b =", a + b);
console.log("a - b =", a - b);
console.log("a * b =", a * b);
console.log("a / b =", a / b);
console.log("a % b =", a % b);
console.log("a ** 2 =", a ** 2);
</SCRIPT>
```

127.0.0.1:5500 says

Enter a:

10

OK

Cancel

127.0.0.1:5500 says

Enter b:

20

OK

Cancel

